Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

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Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

[12pt]article amsmath, amssymb, amsthm

definition Definition Proposition Theorem

Math 250 Proof Portfolio Your Name December 4, 2019

Definitions

An integer x is *even* if $x = 2a$ for some integer a.

A *rectangle* is a quadrilateral all of whose angles are right angles. A *square* is a rectangle all of whose sides are congruent.

1 Proof by cases

If X is a square, then X is a rectangle.

Suppose X is a square.

I'm going to demonstrate some of the cool ways that LaTeXcan format mathematical expressions! Let s be the side length of X and A be the area of X.

Then we have $A = s^2$. Now suppose that s is an even integer. By definition, this means that $s = 2a$ for some $a \in Z$. Then the area of X is $\mathrm{A} = \mathrm{s}^2 = (2a)^2 = 4a^2$, which we also could have formatted over multiple lines as $\mathrm{A} = \mathrm{s}^2$
$= (2a)^2$
$= 4a^2$, which shows that A is a multiple of 4, i.e., $A \equiv 0 \pmod 4$.

We can format set-builder notation like this: $\{\ 2\mathrm{n} : \mathrm{n} \in Z\} \subseteq \{m \in Z : 4|m\}$.

Now suppose that the side length of X is a function of t, where $t \in R$. Then by the chain rule, the derivative of the area is $\frac{dA}{dt} = 2s\frac{ds}{dt}$, which we could also have formatted as $\dfrac{dA}{dt} = 2s\dfrac{ds}{dt}$ if we wanted to make the fractions larger. The net change in the area from $t = t_1$ to $t = t_2$ is $\mathrm{A}(\mathrm{t}_1) - A(t_2) = \int_{t_1}^{t_2} \frac{dA}{dt}\, dt$. Notice how I put a period at the end of that sentence!

I said this was a proof by cases, so let's see how to format lists. Either X is small or X is large. Let's consider those two cases separately:

1. Suppose X is small ...
2. Suppose X is large ...

Here's how to create a bulleted list:

- *Case 1:* Suppose X is small ...
- *Case 2:* Suppose X is large ...

By the way, since a square is by definition a rectangle all of whose sides are congruent, X is also a rectangle. This completes the proof. Marvel at how LaTeXwill create the "end-of-proof-box" automatically:

2 Congruence modulo n

3 Proof by contrapositive

4 Proof by contradiction

5 Induction

6 A proof that a function is bijective

7 The triangle inequality

8 The Pythagorean theorem

Use brackets to give a theorem a name in parentheses: [Pythagorean theorem] For every right triangle with legs of lengths a and b and hypotenuse of length c, we have that $a^2 + b^2 = c^2$.

Compare to: This theorem does not have a name in parentheses since I did not use the brackets.

You could cite a source by saying something like: The idea of the following proof, which is originally due to Euclid, has been borrowed from [2].

Here's the proof.

9 A proof not from this course

Pustaka

[1] R. Hammack, *Book of Proof*. Third edition. 2018.

[2] Wikipedia contributors, *Wikipedia*, "Pythagorean theorem". https://en.wikipedia.org/w/index.php?title= Pythagorean_theorem &oldid=918079404. Online; accessed October 15, 2019.